

HARDWARE AND COMPONENTS BASICS

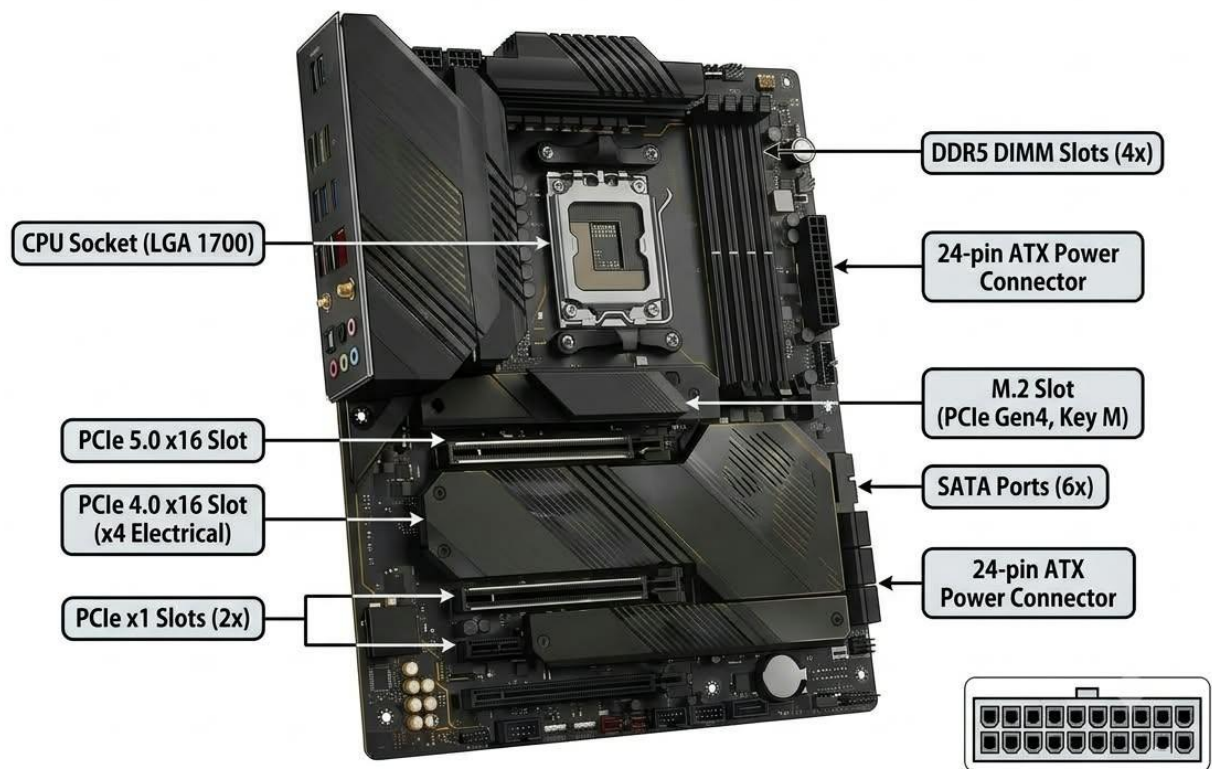
DAY 2 – MOTHERBOARD AND CPU

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1. Labeled ATX Motherboard Diagram

The diagram below shows the main motherboard slots, ports, connectors, and chipset areas.

ATX MOTHERBOARD LAYOUT



*Representative Diagram for Educational Use

Labels: CPU Socket, RAM/DIMM Slots, PCIe Expansion Slots, SATA Ports, USB Headers, M.2 Slot, 24-Pin ATX Power Connector, CPU Power Connector, Chipset, and Rear I/O Ports.

2. Popular CPU Socket Types and Compatible CPUs

A CPU socket is the physical interface between the processor and the motherboard. The CPU and motherboard must support the same socket and platform requirements.

CPU Socket	Example Compatible CPU	Manufacturer
LGA 1151	Intel Core i7-8700K	Intel
AM4	AMD Ryzen 5 3600	AMD
LGA 1200	Intel Core i5-10600K	Intel

Note: Socket compatibility is not the only requirement. The motherboard chipset, BIOS support, and CPU generation should also be checked.

3. Functions of the CPU and Chipset

CPU (Central Processing Unit): The CPU is the main processing unit of a computer. It executes instructions, performs calculations, processes application tasks, and coordinates many operations.

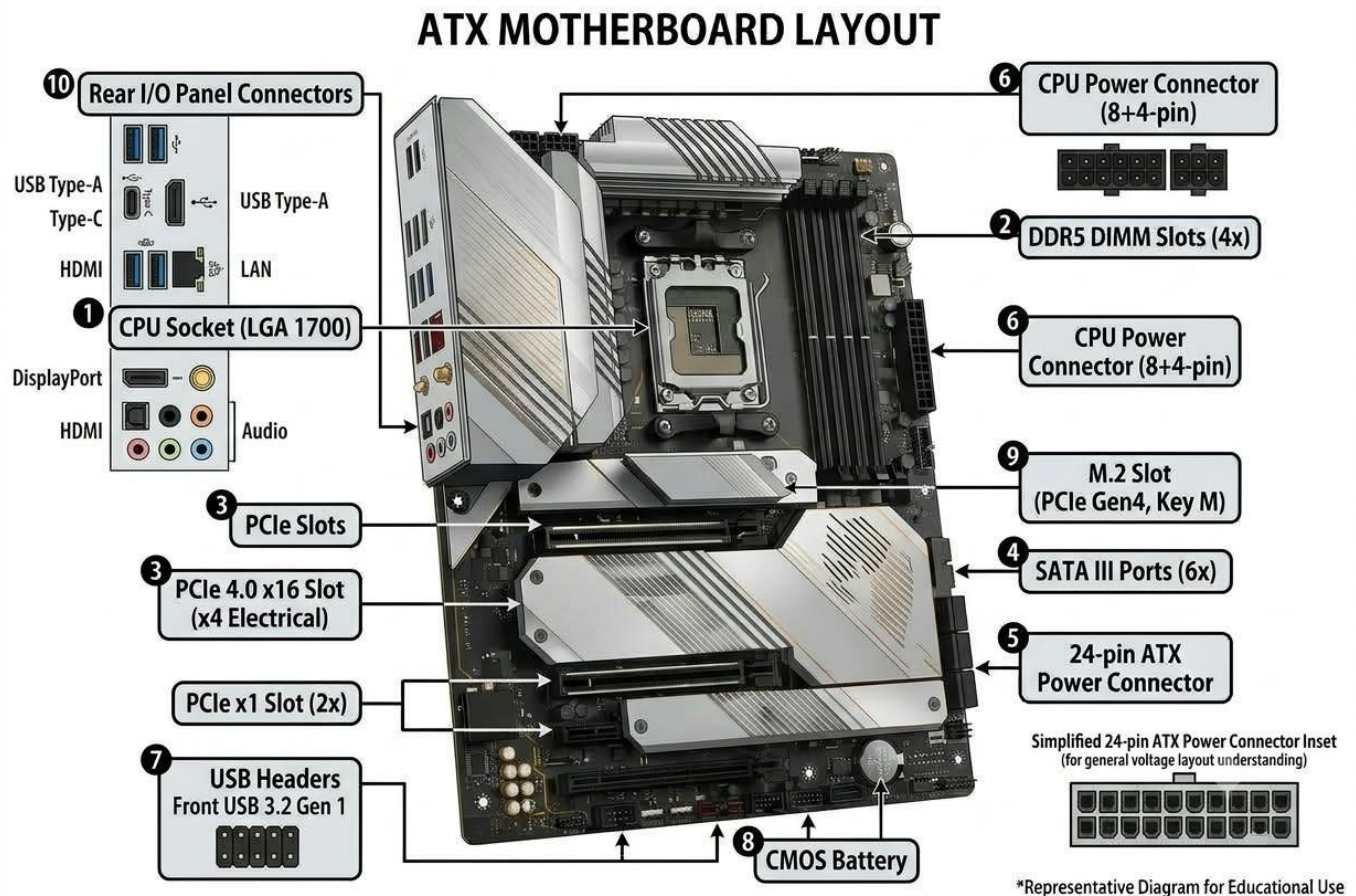
Chipset: The chipset is a group of controller functions on the motherboard that helps manage communication between the CPU and other system components, including storage, USB, expansion, and other motherboard features.

How they work together when opening Spotify:

When Spotify is opened, the operating system loads the application's instructions and required data. The CPU executes those instructions and performs the processing needed to start the app. The motherboard platform and chipset provide the connections and communication pathways used by storage, networking, USB devices, and other hardware. The CPU continues processing Spotify's tasks while the hardware handles input/output, allowing the application to run and play music.

4. Real Motherboard Photo – Identification

The image below is a clear top-down motherboard view prepared for the practical task. Seven different slots/ports are circled and labeled.



Identified parts: 1. CPU Socket 2. RAM/DIMM Slots 3. PCIe Slot 4. SATA Ports 5. 24-Pin ATX Power 6. M.2 Slot 7. Rear I/O Ports

Conclusion

This assignment helped me understand the major parts of an ATX motherboard, common CPU socket types, and the basic roles of the CPU and chipset. I also learned how to identify common motherboard slots, ports, and connectors used in a desktop computer.